

Implementation brief: sports_picks review remediation

Hand this to a coding agent with full access to the sports_picks repository. It is self-contained: background, what to change, acceptance criteria for each change, how to measure success, and what not to do.

1. Background

sports_picks is a personal sports-betting research app (no real money; friends paper-trade). Backend: Python, FastAPI, SQLAlchemy, one SQLite file, APScheduler. Frontend: React 19 + Vite + TypeScript. Data: The Odds API (20,000 credits/month, ~600/day, 2,000 held in reserve), ESPN, MLB Stats API, UFC/boxing scrapers. Runs on the owner's Windows laptop. ~8,300 lines, ~1,560 tests.

An evidence review (2026-09-28) found:

- **The game model adds nothing to the market price.** Best held-out blend of model + market probability put 0.00 weight on the model. Team-level ratings (NFL EPA, 202,851 plays) were 0.88 RMSE worse than the closing spread. Graded record: 217-252 (46.3%), +1.85u at 1u flat — break-even, statistically indistinguishable from zero.
- **Stars don't rank picks.** 5★ 31-29 (51.7%), 4★ 24-28 (46.2%), 3★ 8-6, 2★ 6-12, 1★ 148-177 (45.5%). Non-monotonic, thin samples, nightly threshold refits on correlated picks.
- **Prop "edges" are a units error.** Reported edge averages 77 points because stat-unit gaps (projected yards vs line) are treated as probability points. Props win 50.5% at ~-110 (-4% ROI). MLB props are emitted but cannot be graded (missing stat columns).
- **MMA is +30.91u on 52 bouts (27-25).** That implies ~59% ROI per bet at an average winning price near +207 with a 52% hit rate — implausible as a real edge; prime suspect is a side-price mismatch in grading. Quarantined until audited.
- **Recurring failure mode is "dead wiring":** a job runs, logs success, and its output is thrown away or never used (72 duplicate prop rows, -121u bankroll bug, overwritten prices, unrecorded email picks). Most fixes have been this shape.
- **Emailed picks** (4 recipients) have only been recorded since 2026-09-28, so the email record is currently unmeasurable. The email ranks by model win probability, shows "Model 59% / Price 56%", and its header shows the all-picks record — all misleading given the findings above.

2. Operating rules (apply to every workstream)

1. **Constraints are fixed:** FastAPI + SQLite + React stay. No new dependencies without a written justification. No real money, ever. Email list stays at four. Odds budget: 20,000/month,

~600/day, 2,000 reserve — every change that adds odds calls must state its daily credit cost and what it displaces.

2. **DB constraints beat app-level checks.** Any invariant that can be a UNIQUE/NOT NULL/CHECK constraint or an idempotent upsert must be one. The 72-duplicate bug must become unrepresentable, not just fixed.
3. **Every workstream ships a wiring test, not just unit tests.** The repo has 1,560 tests and the bugs shipped anyway. Each change needs an end-to-end test on a frozen fixture asserting inputs → stored outputs → consumed outputs (e.g., scout output rows are exactly the rows the digest reads).
4. **Frozen rules, pre-registered tests.** Any threshold, cutoff, or rule ships with its definition frozen in config and a pre-registered out-of-sample evaluation plan. No refitting cutoffs on small samples (see §4.3).
5. **Effective sample size discipline.** Picks on the same game are correlated. Every reported aggregate must state raw n and, where feasible, an effective- n estimate. No rule ships on $< \sim 200$ effective picks without an explicit exception signed off by the owner.
6. **Start each workstream with recon.** You cannot see prior review notes beyond this brief — map the relevant modules first (pipeline jobs, grading, email selection, staking, prop pipeline, paper-trading ledger), then change them.
7. **Decided by the owner (2026-09-28, updated same day):** enabled sports are **NFL, MLB, and UFC**; all others (NBA, NCAAB, NCAAF, boxing) default to disabled — see §10 H.1. **Planned rotation:** when the NBA regular season starts (opening night Tue Oct 20, 2026), disable MLB and enable NBA. Still to confirm with the owner: the exact email send-bar numbers (§8 — defaults proposed below), and whether stars are hidden vs labeled experimental (§4.3).

3. Workstream A — Edge computation: shrink to the market

Objective: Replace raw `model_prob - market_prob` edges with shrunk edges. Remove the 20-point refusal (fitted to ~ 100 correlated picks — overfit) and replace it with a principled cap.

Changes:

- A.1. Compute the held-out blend weight of model vs de-vigged market probability per sport (log-loss or Brier, time-ordered splits, no leakage). Default the live edge to the blended probability minus market probability. If the weight is ~ 0 (as measured), the blended probability $\approx$ market and edges collapse toward zero — that is the correct behavior, not a bug.
- A.2. Delete the 20-point edge refusal. Replace with a cap derived from calibration bins (e.g., $\max |edge|$ at the largest bin where predicted vs observed stays within tolerance), computed on held-out data only, frozen in config.
- A.3. Keep the email's +150 price ceiling and the model's underdog ceiling staying off (do not touch).

Acceptance criteria:

- A test with a synthetic history where the model is pure noise: blended weight ≈ 0 and emitted edges ≈ 0 (no picks clear the send bar).
- A test with a synthetic history where the model has real signal: weight > 0 and edges rank picks out-of-sample.
- The old 20-point refusal code path is gone; the cap value lives in config with the calibration evidence that set it (source query + date in a comment or doc).

Measurement: Track per-pick closing-line value (CLV) and ROI on picks generated after this change vs the old regime, labeled by regime. Do not re-enable any volume or staking increase until CLV > 0 over a real sample or held-out weight > 0 .

4. Workstream B — Stars, staking, and selection

Objective: Stars must mean "validated expected-value bucket" or they must not be shown. Staking must not pretend at precision that doesn't exist.

Changes:

- B.1. **Staking** → **flat 1u** for all generated and emailed picks. Keep the quarter-Kelly code path behind a config flag, default off. Re-enable only when: (a) held-out blend weight > 0 , or (b) CLV > 0 over ≥ 300 effective picks. Document the re-enable criterion in config.
- B.2. **Delete the 03:00 nightly recalibration of star thresholds.** Replace with: thresholds frozen in config; re-estimation allowed at most monthly, only on ≥ 300 effective new picks per sport, and only if the new thresholds improve out-of-sample ranking on a pre-registered split. Until a star definition passes the test below: either hide stars from the UI/email or label them "model confidence (experimental)" — owner's call (§2.7).
- B.3. New star definition (to be validated before display): pre-registered, frozen buckets of *shrunk* edge that are monotonic in realized ROI/CLV out-of-sample, with effective-n reported per bucket.

Acceptance criteria:

- Config flag `staking.mode=flat` is default; a test asserts every emitted pick gets stake exactly 1.0.
- Nightly threshold-refit job is removed or gated; a test asserts thresholds are byte-identical across two consecutive runs on new data.
- A validation script exists that takes frozen bucket definitions + out-of-sample picks and reports monotonicity, per-bucket n/effective-n, and a pass/fail — and it is run in CI on fixture data.

Measurement: Out-of-sample bucket monotonicity; star 5★ vs 1★ ROI gap with confidence interval. If it doesn't rank, stars stay hidden.

5. Workstream C — Props: real probabilities or nothing

Objective: A prop "edge" must be a probability difference, and every emitted prop must be gradeable.

Changes:

- C.1. Rebuild the prop pipeline: projection (keep the 40/60 season/form blend as the *mean*) → distribution using historical game-log variance (you already collect box scores) → $P(\text{cover}) = 1 - \text{CDF}(\text{line})$ → edge = $P(\text{cover}) - \text{de-vigged implied probability of the line}$. Account for pushes where the market does.
- C.2. Delete the old "edge" computation and any display of it. The email props section stays unranked and edge-free until C.1 validates.
- C.3. **Grading coverage = 100%**. MLB is an enabled sport (owner decision, 2026-09-28), so this is now blocking: add the missing MLB stat columns (hits, total bases, strikeouts, ...) *or* stop generating MLB props — there is no third option. Any prop market that cannot be graded must not be emitted — enforce with a test, not a comment.
- C.4. Keep the dedup fix (idempotent upsert on the natural key); add the DB unique constraint (§2.2) so the 72-row bug is unrepresentable.

Acceptance criteria:

- ☐ On historical props, $P(\text{cover})$ calibration by decile is reported by a script; worst decile miscalibration is visible, not hidden.
- ☐ A test asserts: for every prop market the generator can emit, a grading path exists (fails if a new market is added without one).
- ☐ Re-running the prop pipeline twice for the same window inserts zero new rows.

Measurement: Prop calibration curve + prop CLV once snapshots exist. Kill any prop market with no measurable edge after 200 effective graded picks.

6. Workstream D — MMA audit and quarantine

Objective: Determine whether +30.91u on 52 bouts is signal or grading bug, before it influences anything.

Changes:

- D.1. Write a one-off audit script: for each graded MMA pick, show side picked, price recorded, opponent price from the snapshot, and result. Flag any pick where the recorded price matches the *opponent's* side or where the price is inconsistent with the snapshot series.
- D.2. Manually review 15–20 flagged/unflagged picks against an independent source. Report: side–price alignment error rate, and the P&L distribution (a few huge hits vs consistent edge).

- D.3. Until the audit passes: exclude UFC/MMA from all headline aggregates, the email header record, and any strategy decision — even though UFC remains enabled for pick generation and grading (owner decision, 2026-09-28), so the audit has live data and the experiment continues cleanly. Add a `quarantined_sports` config list; the UI/email must respect it. Boxing is disabled entirely (§10 H.1); if its grading shares the combat-sports code path, the D.1 audit still covers that path.

Acceptance criteria:

- Audit script exists and runs; its output is a CSV a human can verify in 30 minutes.
- With MMA in `quarantined_sports`, headline record, email header, and leaderboard benchmark all exclude it (test asserts).

Measurement: If the audit clears it, MMA becomes its own isolated experiment with pre-registered evaluation (it would then be the only working strategy and deserves focused study). If it's a bug, fix the grading and restate history.

7. Workstream E — Dead-wiring invariants

Objective: Make "ran, logged success, output unused" impossible or immediately loud.

Changes:

- E.1. **Run manifests.** Every scheduled job writes a manifest row: job name, window, inputs read (counts by source), outputs written (counts by table), downstream consumer, and a consumer-acknowledgment (e.g., digest manifest references the scout manifest it consumed). Add a daily check: any manifest without a matching consumer acknowledgment within its SLA alerts.
- E.2. **Reconciliation assertions** (fail loudly, page the owner): picks generated == picks stored per run; emailed pick IDs $\subseteq$ today's generated pick IDs; email subject counts == body counts; no duplicate (game, market, strategy) rows.
- E.3. **Grading closure job** (daily): every pick with start time older than a buffer (e.g., 12h) must be in a terminal state (graded / void / push). Anything else → alert with the pick IDs. This also catches postponements.
- E.4. **Bankroll conservation** (paper trading): assert `sum(balances) + open_exposure + sum(settled_pnl) == constant` after every settlement run. Alert on mismatch.
- E.5. **CLV depth gate:** the CLV report refuses to compute when any pick's snapshot series has `depth < 2` — it prints "insufficient data," not numbers.
- E.6. **Push/void handling audit:** verify spread/total pushes return stake and postponed/cancelled games void (exclude from records, don't leave ungraded). Add explicit `push` and `void` grade states if missing.

Acceptance criteria:

- Each invariant above has a test that injects the historical failure (duplicate rows, unbalanced ledger, depth-1 CLV, ungraded stale pick, unsettled parlay) and asserts the alert/constraint fires.
- DB migration adds the unique constraints from §2.2.

Measurement: Count of invariant alerts per week — target is zero after the first cleanup pass, and every alert must trace to a real bug or a tuned threshold.

8. Workstream F — Email: honest record, real bar

Objective: The email shows only what readers can follow, gated by a frozen bar, with uncertainty stated plainly.

Changes:

- F.1. **Header shows the emailed record** per sport (from the sent-snapshot grades, §1 background), with n and a plain-language tag: <100 picks "early days — not yet meaningful"; 100–300 "taking shape"; 300+ show the 90% range. Never the all-picks record.
- F.2. **Send bar (defaults; owner confirms):** shrunk edge $\geq 3pp$, price within $-150...+150$, ≤ 3 game picks per sport, ≤ 5 props per sport. Rank by shrunk edge (not model win probability). If nothing clears the bar: send a one-line "no qualifying picks today" email (or nothing — owner's call), and log the reason. Freeze the bar in config; changes require a dated note.
- F.3. **Delete "Model X% / Price Y%" lines.** Replace with one line per pick: side + price + one-sentence rationale in words (keep the existing template strings, shortened). Show CLV ("beat the close by 3¢") once measurable.
- F.4. Keep: +150 ceiling, separate unranked edge-free props section, "research, not betting advice" footer, sent-only-when-something + health check.
- F.5. Grade emailed picks from the sent snapshot (already done); add a "price at send; lines move" disclaimer line.

Acceptance criteria:

- A test constructs a slate where nothing clears the bar and asserts the email is the empty-day variant (or unsent, per owner choice) with the reason logged.
- A test asserts header numbers equal the emailed-picks table, not the all-picks table.
- No email template contains model-vs-price percentage pairs (grep test).

Measurement: Emailed-pick ROI and CLV vs all-pick ROI; send rate; recipient replies as a rough proxy. Evaluate the bar itself after 300+ effective emailed picks.

9. Workstream G — UI/UX: the honest dashboard

Objective: Friends compare themselves against the model without mistaking noise for skill.

Changes:

- G.1. **Emailed-picks view becomes the default Track Record tab** (keep the planned week/month/star breakdowns, 90% ranges, break-even rate, drawdown, losing streaks; game picks and props separate). All-picks view moves one click down, labeled as the research firehose.
- G.2. **Leaderboard ranks by shrunk ROI** (pulled toward the vig-implied mean; document the shrinkage constant), 10-bet minimum stays as display-only, raw ROI and units shown beside the rank. Model appears as a player at 1u flat per emailed pick, unranked under 10 bets.
- G.3. **Parlays segregated**: separate parlay leaderboard (or excluded from the main board) since they multiply the vig; fix settlement first — parlays must settle and payouts must reach balances before any public link (already found in planning; this is stop-ship).
- G.4. Annotate aggregates with effective-n ("312 picks, ~190 effective — many share games"). Add per-pick CLV display when snapshots allow.
- G.5. **Toolbar pipeline refresh**: show the estimated odds-credit cost and require confirmation before spending. Never one tap → budget burn.
- G.6. **Before any public link**: per-player PIN, owner key for admin, bets rejected on finished games. Cloudflare tunnel is the only ingress.

Acceptance criteria:

- ☐ Tests assert the default tab's numbers come from the emailed-picks table.
- ☐ A test with a 7-3 short-odds bettor vs a 40-30 bettor asserts the shrunk ranking doesn't crown the 7-3 start.
- ☐ A test places a parlay on a finished game and asserts rejection; a test settles a parlay and asserts the balance updates.

Measurement: Friend behavior proxy — fewer "hot week = skill" misreads is qualitative; the quantitative check is that leaderboard rank correlates with CLV, not with bet count.

10. Workstream H — Narrow the surface, predict movement

Objective: Fewer markets, deeper; and repoint the model at the one prediction the evidence doesn't rule out.

Changes:

- H.1. **Per-sport kill switches** in config. Defaults per owner decision (2026-09-28, updated same day): **enabled = NFL, MLB, UFC; disabled = NBA, NCAAB, NCAAF, boxing**. Disabled sports generate nothing, fetch no odds, and cost no credits. Everything downstream (email, UI, grading closure) must handle a disabled sport without alerts. **Seasonal rotation:** on/around NBA opening night (Tue Oct 20, 2026), disable MLB and enable NBA — a single config change; the agent must verify the swap took effect via the credit-counter test. Note:

UFC stays enabled for collection/grading but remains quarantined from headlines pending the \$6 audit.

- H.2. **Volume cap:** only the top shrunk-edge decile is ever staked/mailed; the rest is research exhaust, clearly labeled in the UI.
- H.3. **Line-movement experiment (separate track, pre-registered):** using opening price, early movement, and cross-book disagreement as features, predict direction of line movement (toward/away). Evaluate on CLV, not win rate. This reuses the pipeline; it does not touch live pick selection until it beats a null on held-out data.
- H.4. **Cheap features with priors only:** wind (football totals), umpire/referee crews (MLB totals, NBA props), rest/travel (NBA). Each gets a one-page pre-registration (hypothesis, data source, evaluation metric, kill criterion: no edge after 200 effective picks → deleted). No new team-level ratings, ever (ruled out by the EPA test).

Acceptance criteria:

- ☐ Disabling a sport in config yields zero odds calls and zero picks for it (test with a credit-counter mock).
- ☐ The movement experiment has a pre-registration doc and a null baseline before it runs.

Measurement: Odds credits/day (must stay $\leq$ ~600 with reserve intact — add per-job credit logging to the run manifest, \$7); CLV of the movement experiment vs null.

11. Explicit non-goals

- Do not widen email distribution. Do not add real-money plumbing. Do not propose always-on hosting (separate later project). Do not add team-level football ratings. Do not re-enable Kelly, nightly refits, or raw model edges without the pre-registered criteria above being met. Do not optimize for win percentage — the target metrics are ROI and CLV.

12. Definition of done (whole brief)

1. All acceptance checkboxes above are green in CI, including the new wiring tests.
2. A `docs/review-remediation.md` records: what changed, the frozen config values and the evidence/dates behind them, and the re-enable criteria for Kelly/stars/volume.
3. Odds-credit burn per day is logged per job and stays within budget with the reserve intact for a full week.
4. The owner can answer: "what is our edge, and what would prove us wrong?" — from the dashboard alone.